

Vesiculobullouse disease

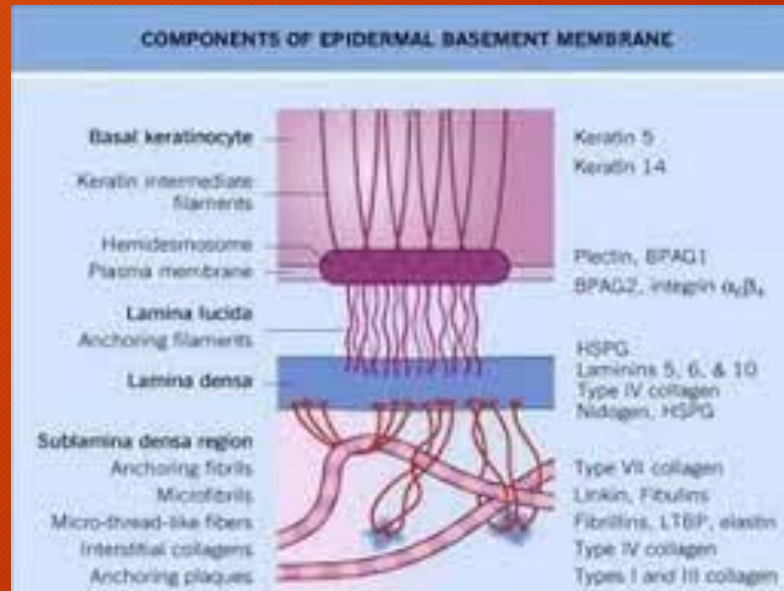
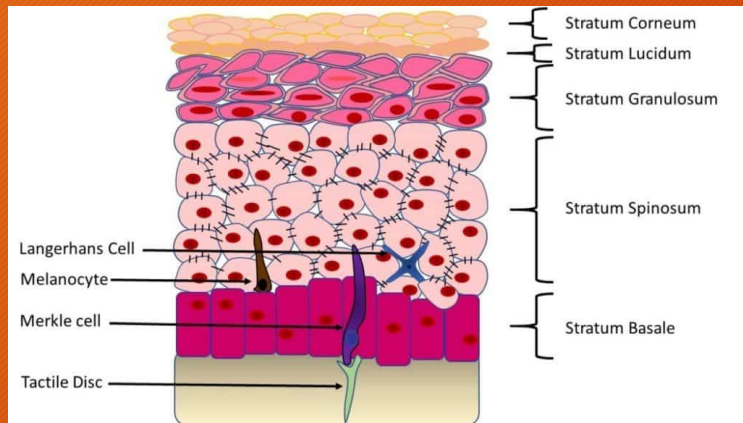
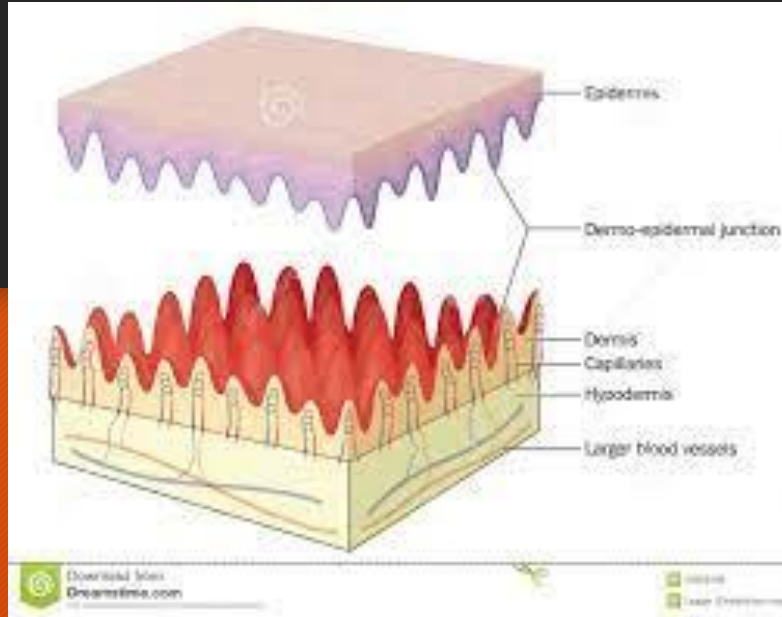
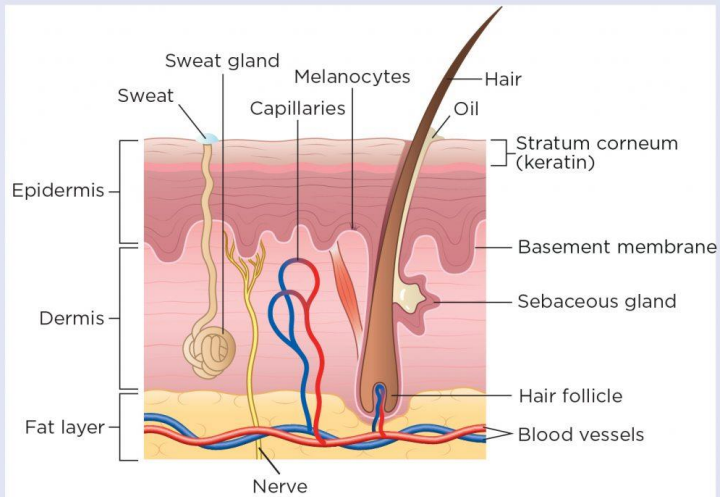
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- Bullous diseases are diseases characterized clinically by blisters i.e. vesicles (if small) and bullae (if large)





Fig 1. Cross-section through the skin



- Blisters in the skin may occur
 1. in the epidermis (called intra-epidermal) due to defects in the adhesion between keartinocytes
- 2. below the epidermis (called sub-epidermal) due to defects in the proteins of the basement membrane zone


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graph TD; A[Cause of bullae formation] --> B[Immunological]; A --> C[Non immunological];
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Cause of
bullae
formation

Immunological

Non
immunological

Immunological

Autoantibodies attack structural proteins in the skin resulting in blisters

Pemphigus
Pemphigoid

Non immunological

Genetic :EB , Darier , Healey,
Healey

Traumatic

Infection :HSV , HZ , impetigo

Inflammatory :eczema

Metabolic :pct

Drug reaction

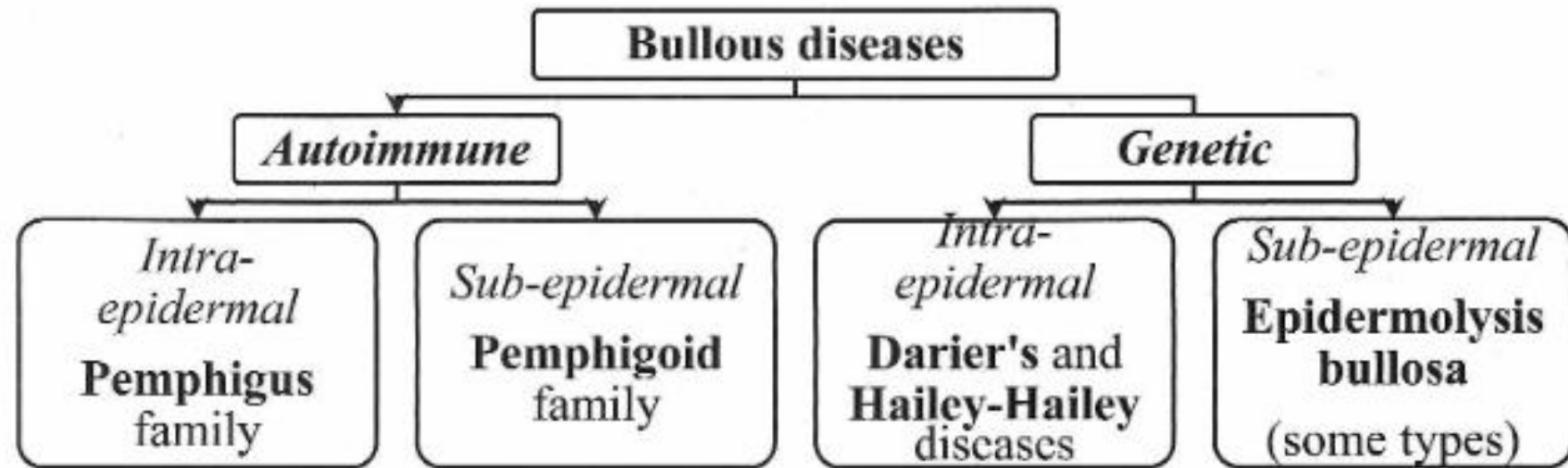


Fig 0.3 Autoimmune and genetic blistering diseases.

Table 0.1 Autoimmune and genetic blistering diseases.

		Level of blister	
		Intra-epidermal	Sub-epidermal
Mechanism of blister formation	Autoimmune	Pemphigus family	Pemphigoid family
	Genetic	Darier's disease Hailey-Hailey disease	Epidermolysis bullosa (some types)

How to diagnose bullous disease

- History taking;
 - Age of onset:
 - Since birth —> EB.
 - Middle age —* pemphigus,
 - Elderly —> pemphigoid.
- - Family history: positive in EB.
 - Relation to trauma —▶ EB.
 - Drug intake: e.g. ACE inhibitors (drug-induced pemphigus).
- - Symptoms:
 - Asymptomatic.
 - Pruritus —> bullous pemphigoid / dermatitis herpetiformis,
 - Painful erosions —▶ oral pemphigus / Hailey-Hailey disease.

- **Examination:**

-

- General examination:

-

- General condition —> poor in oral pemphigus.
G Associated symptoms / systemic affection

- Skin: examine the blisters for

-

- o Flaccid / tense,

- o Small / large.

- o On normal / erythematous skin,

- o Contents (clear fluid / blood / pus),

- o Distribution (site and symmetry),

- o Configuration e.g. annular in IgA pemphigus,

- o Healed blisters (scarring / post-inflammatory hyper- or hypopigmentation).

- Mucosa: oral, conjunctival, nasal, genital
- - Others: e.g. nail and teeth affection

- **Investigations:**

-

- **Detection of skin blister:**

- o Skin biopsy for routine histopathology.
 - o Cytologic examination (Tzanck smear).

-

- **Detection of autoantibodies (in autoimmune bullous diseases):**

- o Immunofluorescence (IF),
 - o Enzyme-linked immunosorbent assay (ELISA),
 - o Immunoprecipitation and immunoblotting.

-

- **Investigations for genetic blistering diseases (i.e. EB):**

- o Transmission electron microscopy (TEM).
 - o Immunofluorescence antigenic mapping

Histeopathology

- Site of biopsy: early small intact vesicle. The whole vesicle should be biopsied.

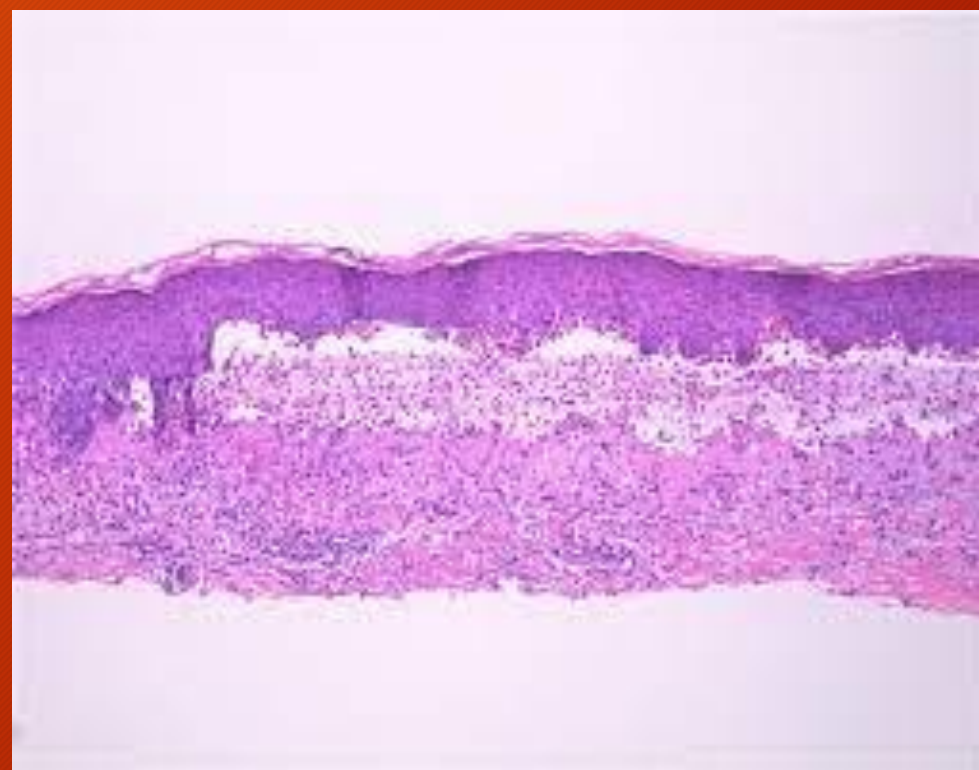
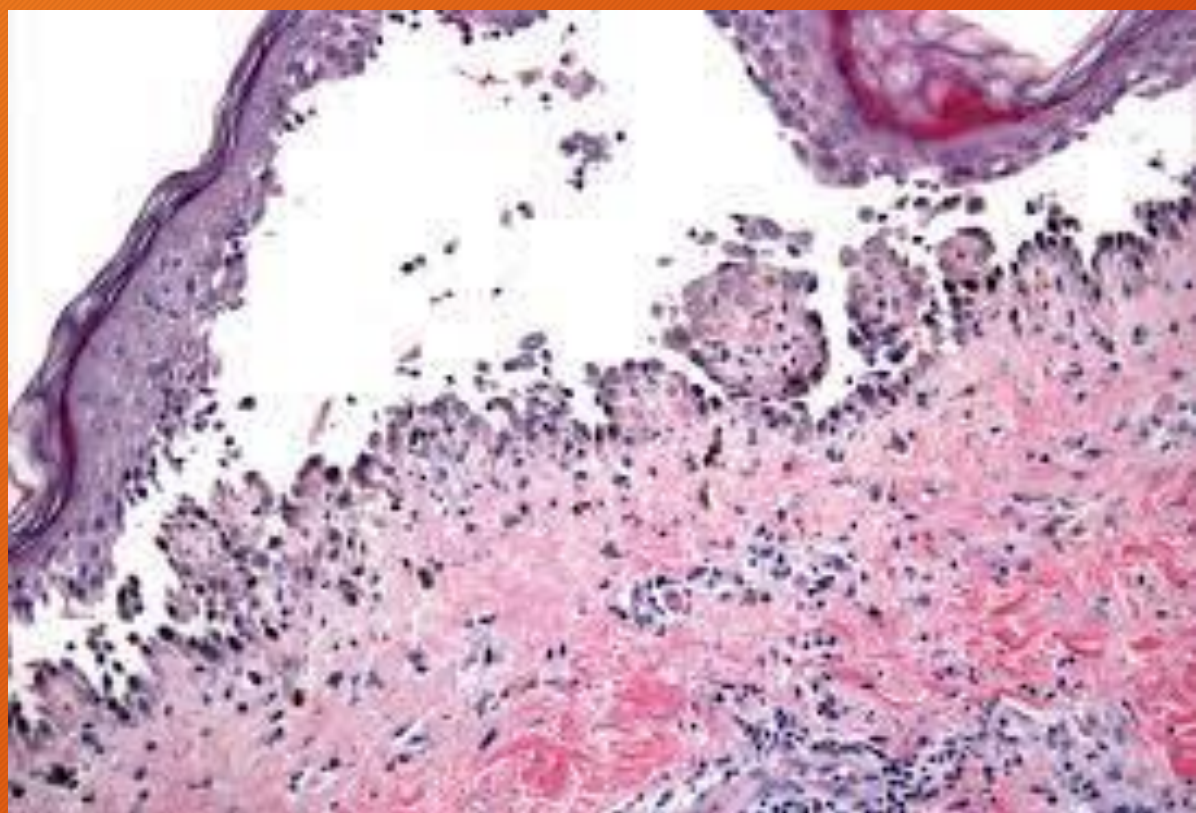
1. the level of the blister

- o intra-epidermal —>■ e.g. pemphigus,
- o sub-epidermal —> e.g. pemphigoid.

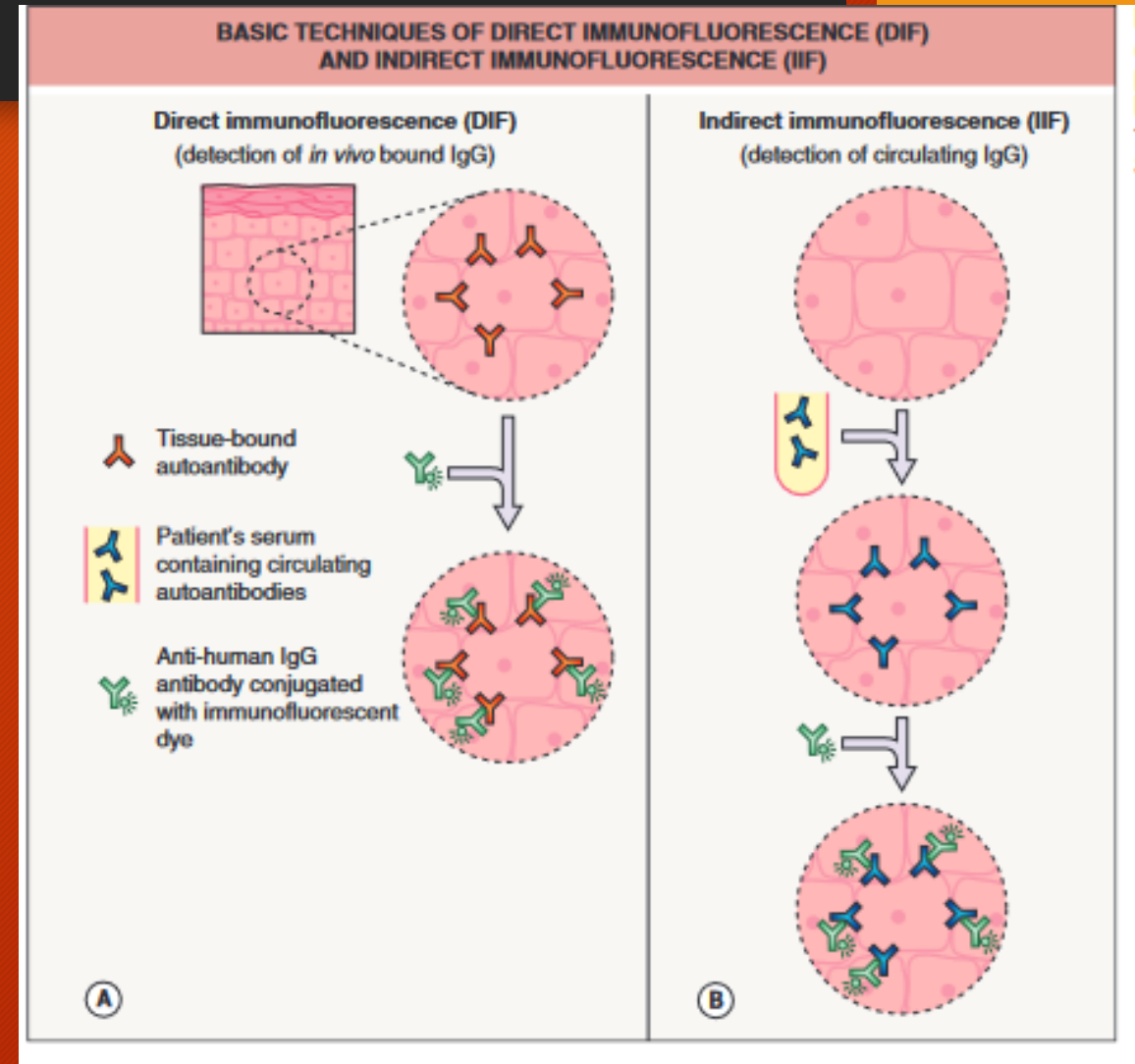
. 2- the mechanism of blister formation e.g. acantholysis —> pemphigus

. 3. the nature of the inflammatory infiltrate
eosinophils, neutrophils or minimal infiltrate.





- Definition: a method to detect pathogenic autoantibodies
- whether they are
 1. bound to their target antigens in the skin :DIF
 2. circulating freely in the blood (before reaching .



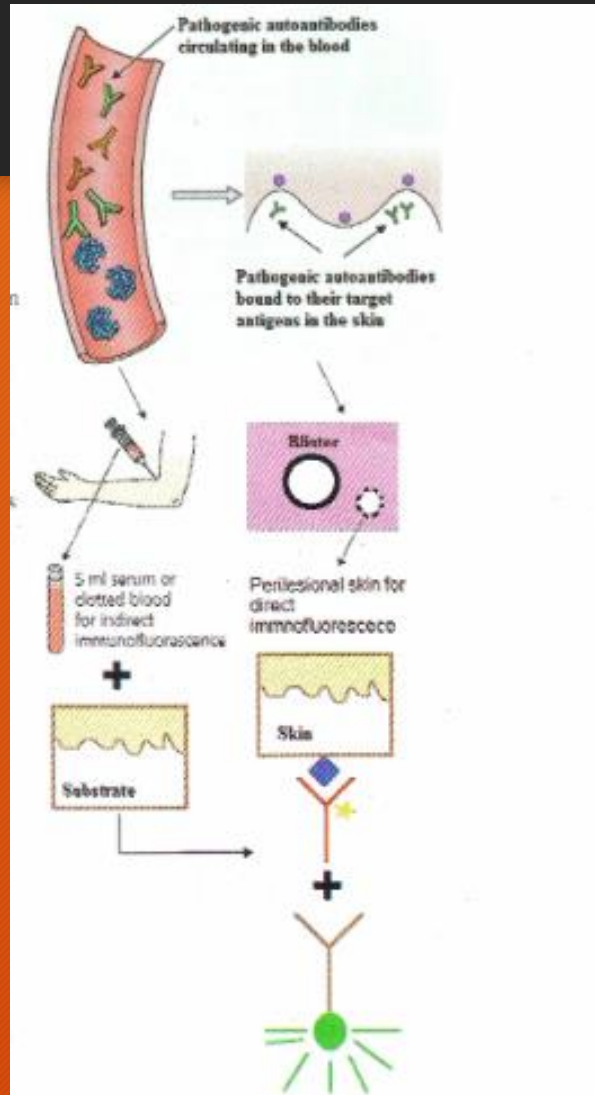
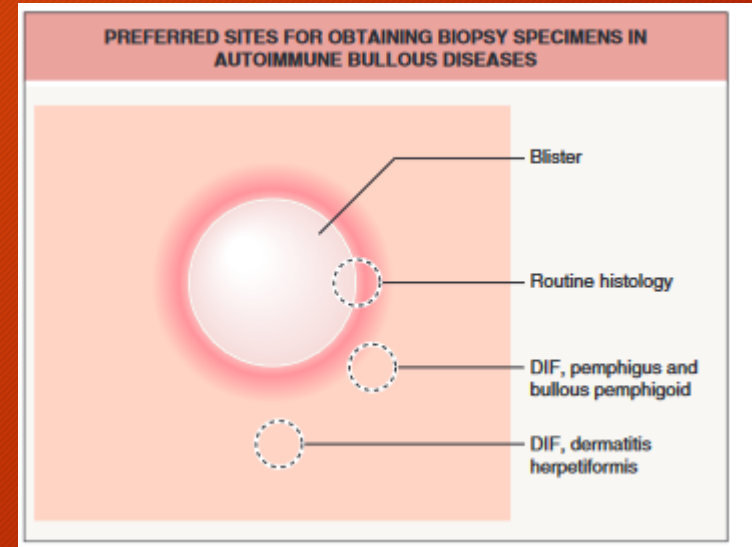


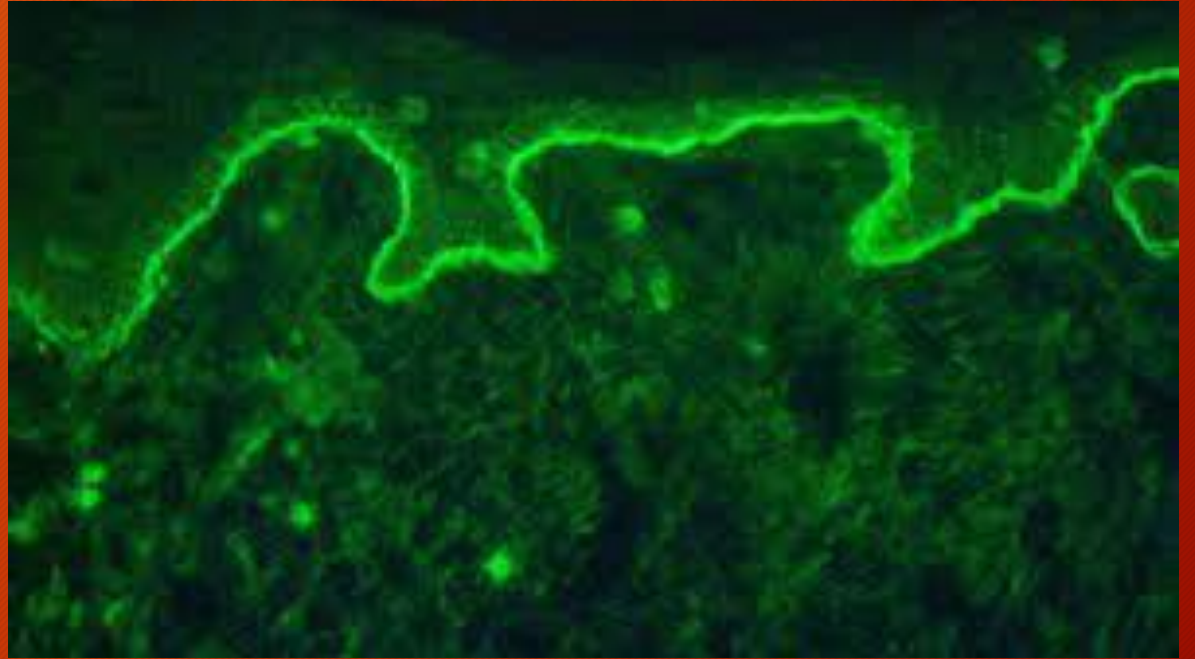
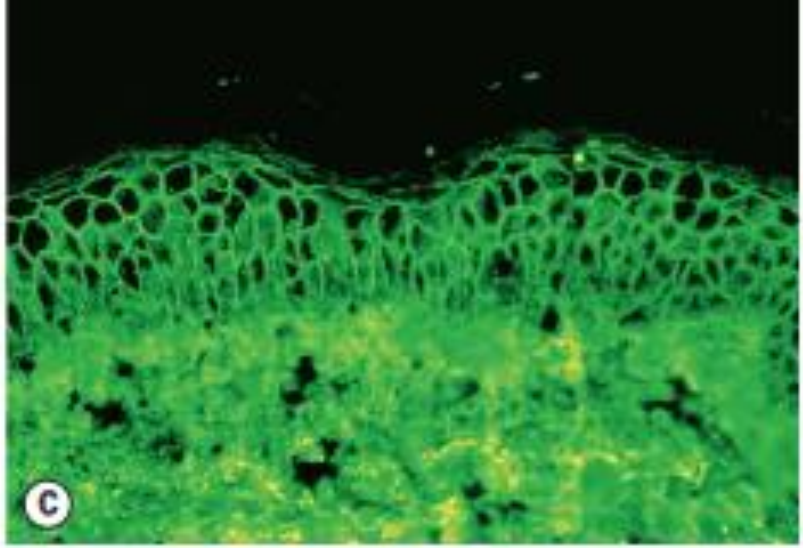
Fig 0.7 Steps of the 2 types of IF test



INDIRECT IMMUNOFLUORESCENCE (IIF) – RECOMMENDED SUBSTRATES

Type of pemphigus	Recommended substrates (autoantibodies)
Pemphigus vulgaris	Monkey esophagus (anti-Dsg3)
Pemphigus foliaceus	Human skin or guinea pig esophagus (anti-Dsg1)
Paraneoplastic pemphigus	Monkey and guinea pig esophagus (anti-Dsg1, anti-Dsg3) Rat bladder (anti-plakin)
Bullous pemphigoid Linear IgA bullous dermatosis	Human skin, salt-split
Mucous membrane (cicatricial) pemphigoid	Human skin, salt-split; normal oral or genital mucosa or conjunctiva

Table 29.5 Indirect immunofluorescence (IIF) for pemphigus and bullous pemphigoid – recommended substrates.



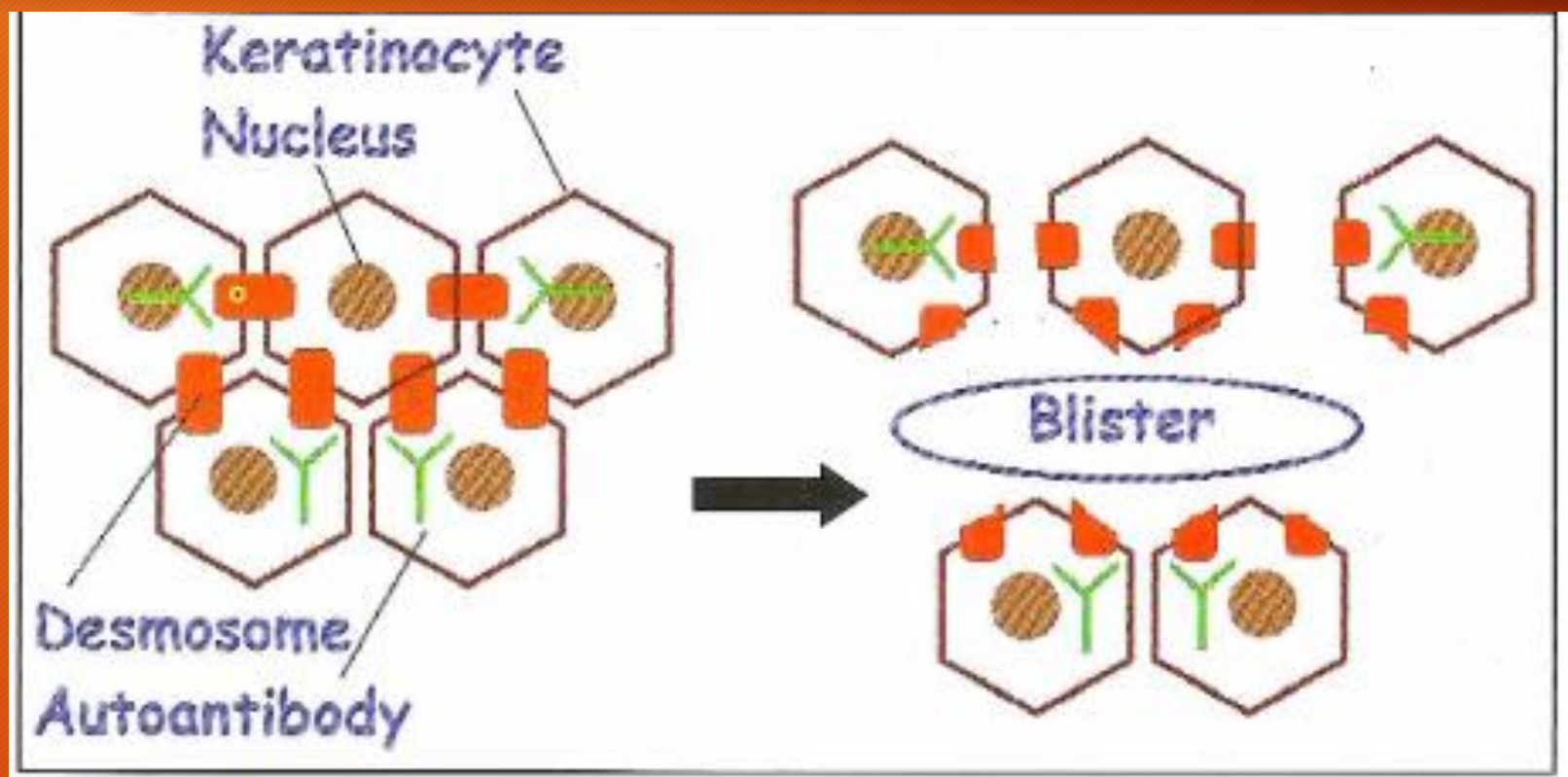
Pemphigus family

CLASSIFICATION OF PEMPHIGUS

- Pemphigus vulgaris
 - Pemphigus vegetans
- Pemphigus foliaceus
 - Pemphigus erythematosus: localized
 - Fogo selvagem: endemic
- Herpetiform pemphigus
- Drug-induced pemphigus*
- Paraneoplastic pemphigus
- IgA pemphigus

*Captopril, penicillamine

Table 29.1 Classification of pemphigus.

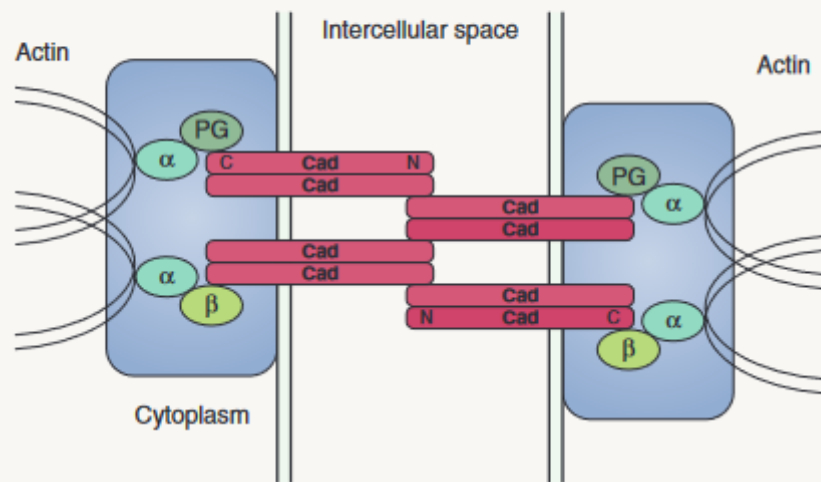


Types	Subtypes	Variants	
1) Pemphigus vulgaris	■ <i>Mucosal-dominant</i> type ■ <i>Muco-cutaneous</i> type	■ Pemphigus <i>vegetans</i> 2 subtypes: ■ Severe $\Rightarrow$ <i>Neumann</i> type ■ Mild $\Rightarrow$ <i>Hallopeau</i> type	■ Herpetiform Pemphigus*
2) Pemphigus foliaceus		■ Localized variant $\Rightarrow$ Pemphigus <i>erythematosus</i> ■ Endemic variant $\Rightarrow$ <i>Fogo selvagem</i>	
3) Paraneoplastic pemphigus (PNP)			
4) IgA pemphigus	■ <i>Sub-corneal pustular dermatosis (SCPD)-like</i> type ■ <i>Intra-epidermal neutrophilic (IEN)</i> type		
5) Drug-induced pemphigus			

***Herpetiform Pemphigus:** a clinical variant of pemphigus foliaceus (most patients) or pemphigus vulgaris (the remainder of cases).

THE ADHERENS JUNCTION AND DESMOSOME

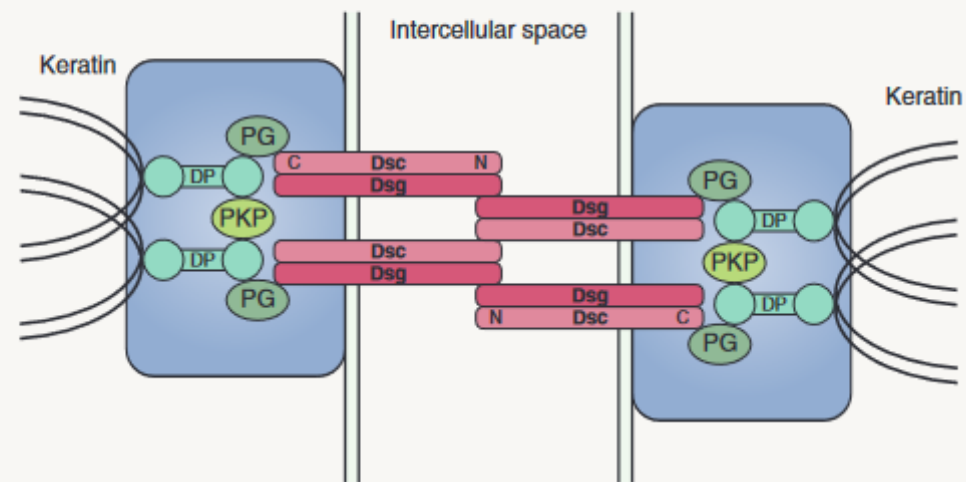
A Adherens junction



Plasma membrane

- Classic cadherin (Cad)
- Plakoglobin (PG)
- β-catenin (β)
- α-catenin (α)

B Desmosome



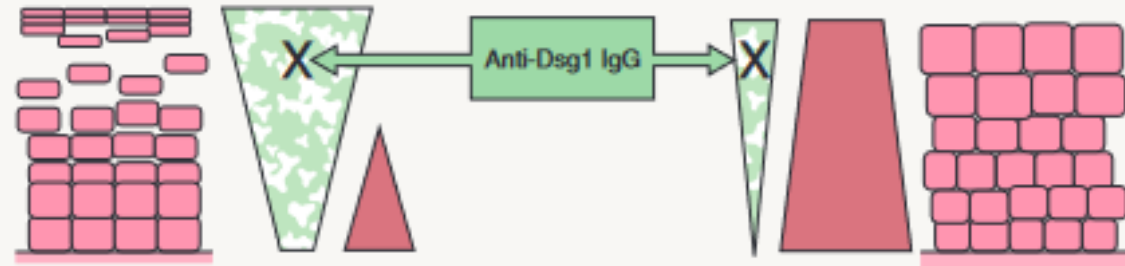
- Desmoglein (Dsg)
- Desmocollin (Dsc)
- Plakoglobin (PG)
- Plakophilin (PKP)
- Desmoplakin (DP)

DESMOGLEIN COMPENSATION THEORY

(A) Skin

(B) Mucous membrane

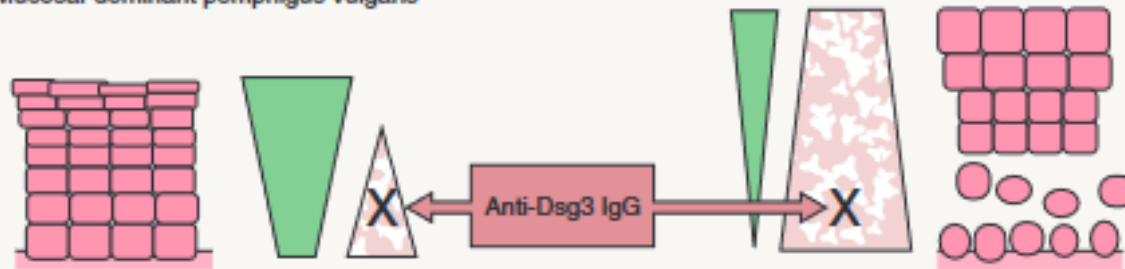
1. Pemphigus foliaceus



Superficial skin blisters

No mucosal lesions

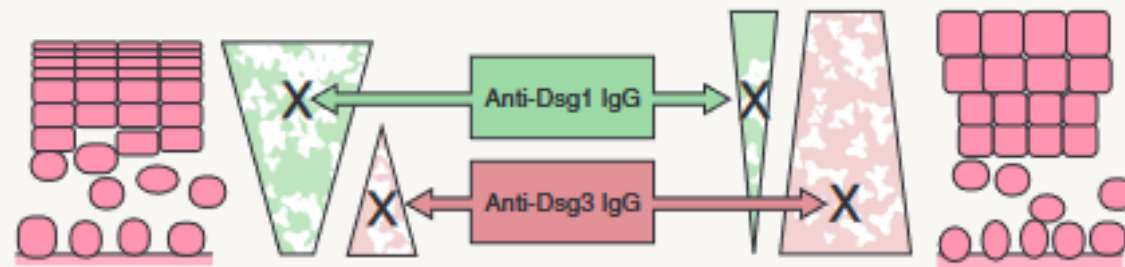
2. Mucosal-dominant pemphigus vulgaris



No or limited skin lesions

Mucosal erosions

3. Mucocutaneous pemphigus vulgaris



Deeper skin blisters

Mucosal erosions

PEMPHIGUS VULGARIS

- Pemphigus vulgaris is the commonest and main type of pemphigus.

It is a chronic autoimmune intra-epidermal blistering disease of **the mucosa** and skin characterized by flaccid blisters and erosions

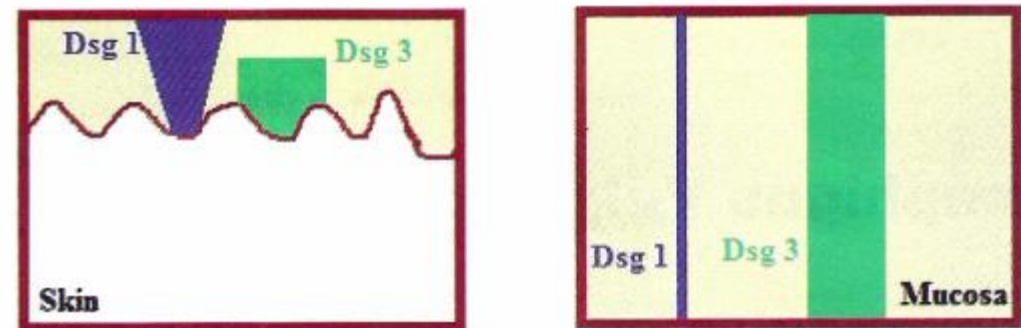
due to autoantibodies attacking the desmosomal proteins.



- Pathogenesis
- (target protein in the desmosome)
 - Mucosal-dominant type: desmoglein 3 (Dsg 3)
 - Muco-cutaneous type: desmoglein 3 (Dsg 3) + desmoglein 1 (Dsg 1).

Compensatory mechanism

- Mucosal-dominant subtype: affection of Dsg3 only
 - In the skin: Dsg1 compensates (NO skin lesions).
 - In the mucosa: Dsg1 CAN'T compensate ^ mucosal lesions
- b) Muco-cutaneous subtype:
 - Affection of BOTH Dsg3 and Dsg1 —► affection of both skin and mucosa



(a) (b)
Fig 2.1 Distribution of Dsg1 and Dsg3 in the skin (a) and mucosa (b).



Fig. 29.5 Pemphigus vulgaris – oral involvement. Essentially all patients develop painful oral mucosal erosions. The most common sites are the buccal (A) and palatine mucosae, but lesions can also develop on the gingivae (B) and tongue (C). A, Courtesy, Lorenzo Cerroni, MD; B,C, Courtesy, Jeffrey P Callen, MD.



- Middle age , equal sex , ashkenaz jews
- General condition : poor
- Symptoms: oral >> painful , skin >> asymptomatic
- Mucosa: **All patients 100%** , painful erosion ,
oral , nasal ,throat , vagina
Conjunctiva , penis



- Skin : 50%-70%
 - flacid thin blister>> easily rupture>pianfull erosion
 - Scalp , face , upper back , chest ,
 - heal without scar
- + ve nikolsky
- +ve asboe Hansen sign

Nikolsky's sign

Easy to peel the skin off



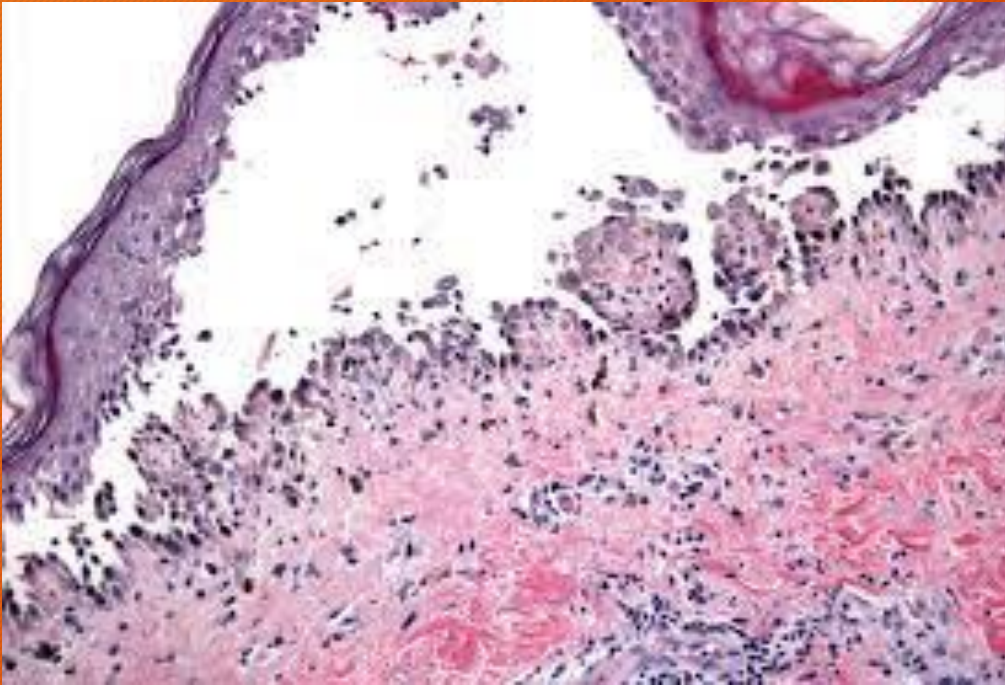
pressure or rubbing with a finger on normally looking Skin will separate epidermis from dermis

gentle pressure on an intact bulla forces the fluid to spread under the skin away from the site of pressure (spread of bullae)

Asboe-Hansen sign

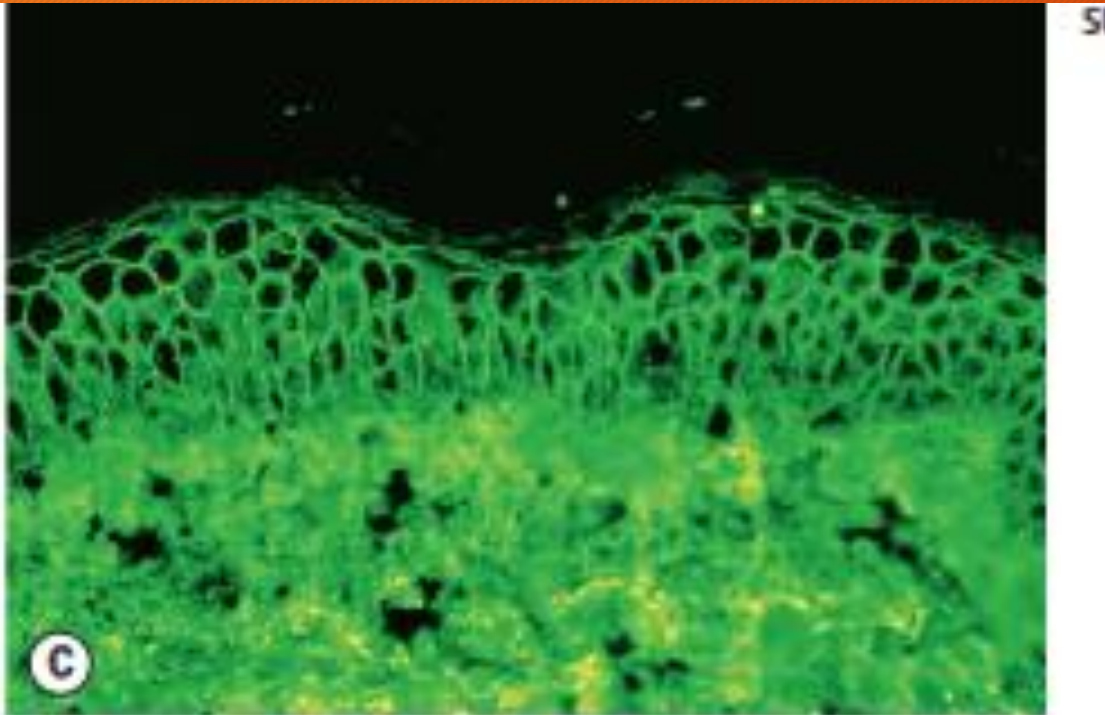


Histeopathology



- 1. Intra-epidermal blister just above the basal cell layer (supra-basilar).
(appearance of a “row of tombstones”)
- 2. Loss of cell-cell adhesion of keratinocytes (acantholysis)
(acantholytic) keratinocytes are seen in the blister cavity (supra-basilar acantholysis)
- 3. Inflammatory cells: eosinophils (in the blister cavity and the dermis).

immunofluorescence



- DIF: IgG $\pm$ C3 on the keratinocyte cell surfaces (inter-cellular) (Fig 2.4). This pattern is referred to as “chicken wire appearance”
- IIF: circulating IgG better detected on Monkey esophagus

DD OF MUCOSAL LESION



1. Acute herpetic stomatitis
2. Aphthous stomatitis
3. Erythema multiforme or Stevens-Johnson syndrome
4. Lichen planus
5. Systemic lupus erythematosus
6. Mucous membrane (cicatricial) pemphigoid

Treatment

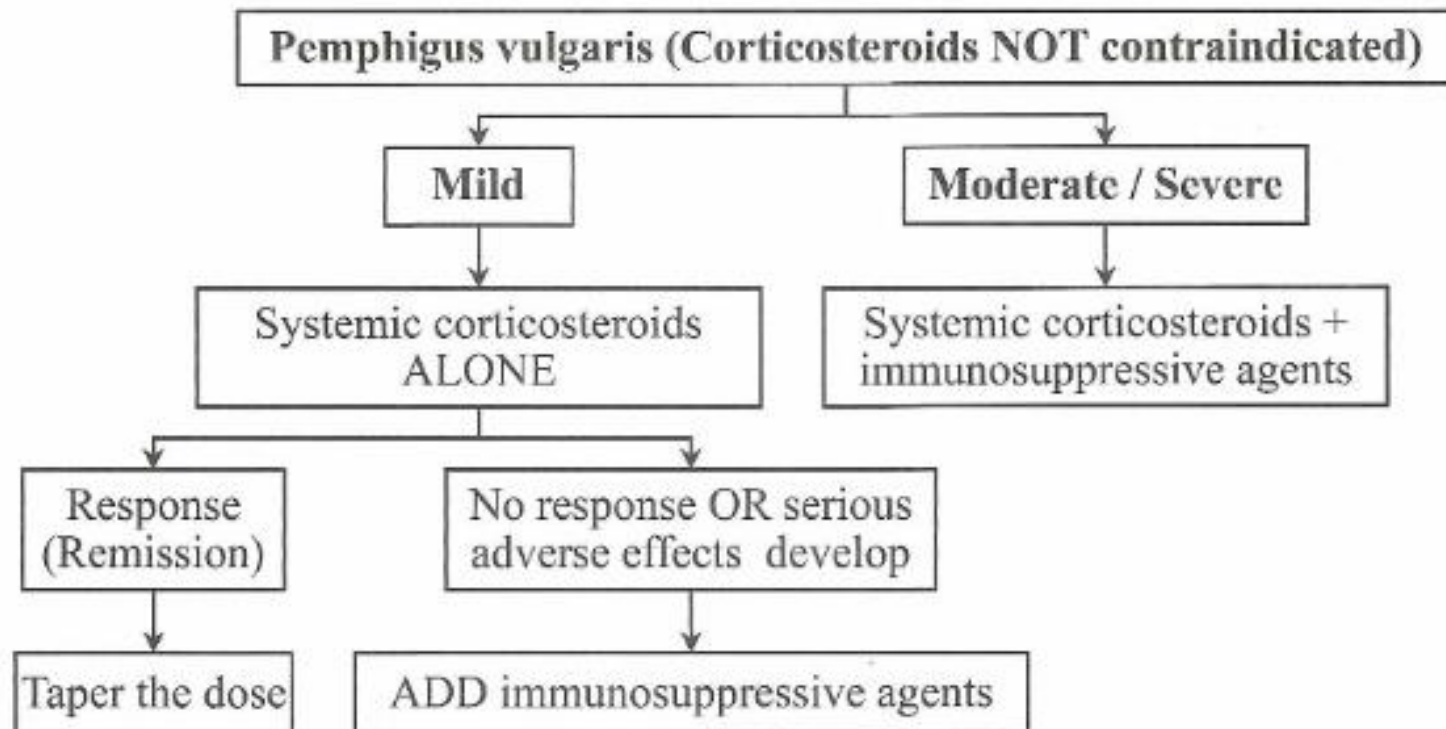


Fig 2.5 Algorithm for treatment of pemphigus vulgaris.

- Systemic treatment

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- **Systemic corticosteroids:**

- The mainstay of therapy for pemphigus

-

- Oral prednisone (standard treatment): 1 mg/kg/day (usually 60 mg/day).
 - Intra-venous pulse therapy: Methylprednisolone 1 gram /day (over a period of 2-3 hours, with continuous cardiac monitoring) for 3-5 consecutive days - Used for severe cases.

-

- Immunosuppressive agents

-

- Used for their corticosteroid-sparing effect to reduce the side effects of the corticosteroids

- Topical treatment
 - Topical corticosteroids (for localized mild lesions. They are rarely effective alone).
 - Topical antibiotics.
 - Topical immunomodulators (e.g. tacrolimus)

<i>Others</i>
High-dose IVIG “intra-venous immunoglobulins” • A blood product prepared from pooled plasma • It has immune-modulatory effects when used in a high dose (400 mg/kg/day for 5 consecutive days) • Effective and safe for steroid-resistant pemphigus
Plasmapheresis • Quickly reducing the titers of circulating autoantibodies • Used for severe pemphigus unresponsive to a combination of prednisone and immunosuppressives (1–2 times per week)
Extracorporeal photopheresis (ECP): 2 days per month

- How to monitor the therapeutic response?
- - Clinically: (1) number of new blisters per day and
 - (2) rate of healing of lesions.
 - If NO new lesions for 2 weeks —► clinical remission.
- Titer of circulating auto-antibodies: determined by indirect IF or ELISA.
- Once clinical remission is obtained, changes in the titer of circulating autoantibodies are helpful in adjusting the dose of prednisone.

THERAPEUTIC LADDER FOR PEMPHIGUS VULGARIS



Standard treatment

Oral prednisone 1 mg/kg/day as an initial dose (usually 60 mg/day) (1)

Aggressive treatment

Immunosuppressive agents in combination with oral prednisone:

Azathioprine 2–4 mg/kg/day (usually 100 to 300 mg/day) (1)

Mycophenolate mofetil 2–3 g/day (2)

Cyclophosphamide 1–3 mg/kg/day (usually 50 to 200 mg/day) (2)

Cyclosporine 3–5 mg/kg/day (2)

Pulse methylprednisolone 1 g/day over a period of 2–3 hours for 3–5 consecutive days (2)

Methotrexate 7.5–20 mg/week (3)

Pulse cyclophosphamide 500–1000 mg/m² every 4 weeks (3)

Plasmapheresis 1–2 times per week, at the onset (2)

High-dose IVIg 400 mg/kg/day for 5 consecutive days (1); may need to be repeated monthly

Rituximab 375 mg/m² once weekly for 4 weeks (2) or 1 g initially then at two weeks; either regimen may need to be repeated every 3–6 months (1)

Extracorporeal photopheresis 2 days per month (3)

Topical treatment

Topical corticosteroids (1), especially Class I to localized persistent sites

Topical antibiotics (2)

Topical immunomodulators (e.g. cyclosporine, tacrolimus) (3)

Intralesional therapy

Corticosteroids (3)

Rituximab (5 mg/cm²) (3)

Pemphiguse vegetans

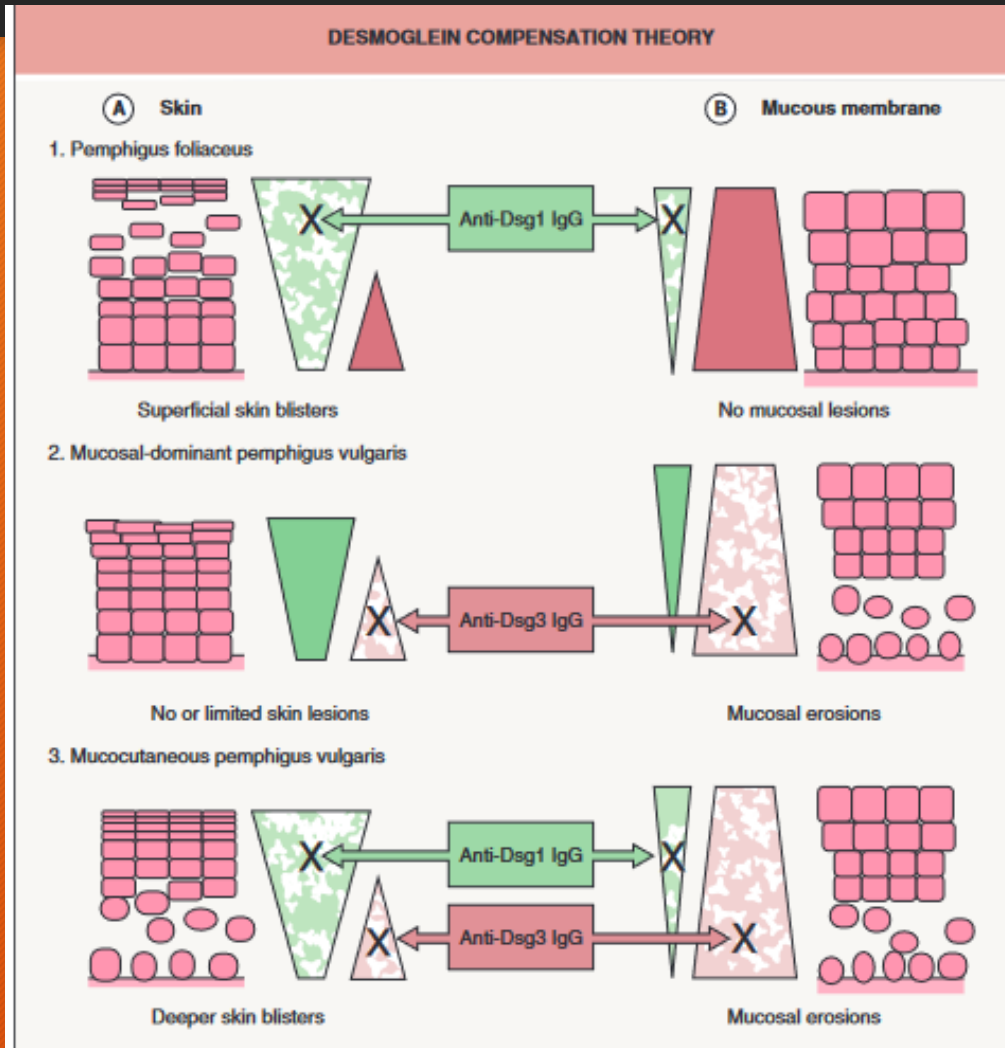


Fig. 29.7 Pemphigus vegetans. Large, thick, vegetating, papillomatous plaques arising in conjunction with erosions. Healed lesions have residual postinflammatory hyperpigmentation.



- Sever :neumanntype
- Mild :hallopeau type
- Flaccid blister >>erosion >>fungoid vegetation
- Intertrigenous area ,scalp , face
- H.P: acanthosis , Papillomatosis, intense inflammatory infiltrate
- DD :heiley heiley , pemphigoid vegetans, pyodermatitis vegetanse

Pemphigus Foliaceus



- Pathogenesis (target protein in desmosome)
 - Desmoglein 1 (Dsg 1)
 - In the skin: Dsg3 functionally compensates for the impaired Dsg1 in the lower part of the epidermis ■ => superficial blisters
 - Desmoglein 3 (Dsg 3)
 - In the mucosa: cell-cell adhesion is mainly mediated by Dsg3 => NO blisters in the mucous membranes

- Middle age , equal sex
- General condition : good
- Symptom e: burning , pain in skin
- No mucous membrane
- Skin : scaly crusted erosions on erythematous base
- Seborrheic distribution
- Erythroderma
- +ve nikolesky sign



Fig. 29.8 Pemphigus foliaceus. A,B Multiple erosions arising on an erythematous base as well as the characteristic scale-crust; the latter can vary in thickness. A common site of involvement is the trunk. C As the disease progresses, the lesions become confluent, but because the vesicles are fragile and rupture easily, only erosions with scale-crust are observed. D The scales have been likened to cornflakes.

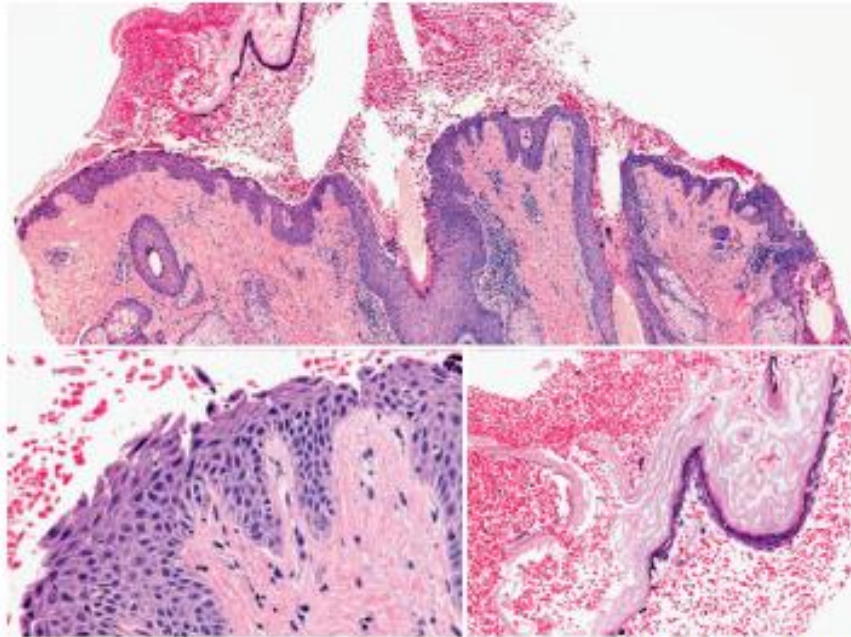
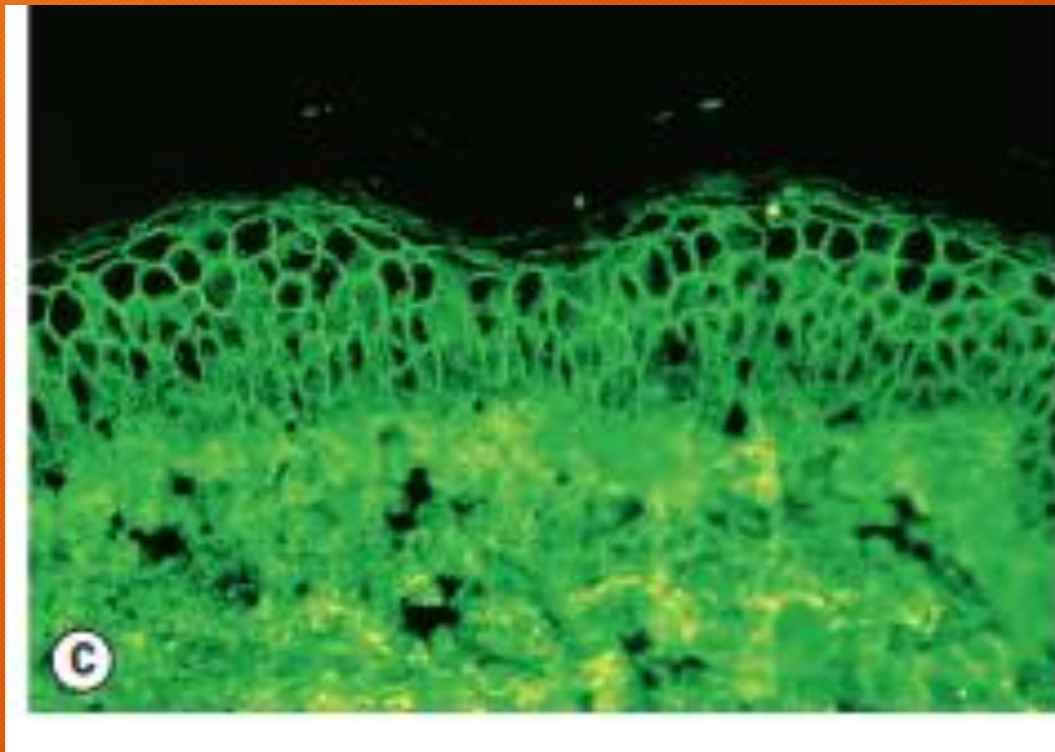


Fig. 29.14 Pemphigus foliaceus – histologic features. Because of the fragility of the blisters, the blister roof may become detached (right inset). The split is in the upper spinous layer of the epidermis and acantholytic cells are readily apparent (left inset). *Courtesy, Lorenzo Cerroni, MD.*

- H.P :Acantholysis in the upper epidermis,
 - Sometimes the blister cavity contains neutrophils.
 - Eosinophilic spongiosis
- ® The dermis shows a moderate number of inflammatory cells



- DIF: IgG $\pm$ C3 on the keratinocyte cell surfaces (inter-cellular)
This pattern is referred to as “chicken wire appearance”.
- IIF: circulating IgG better detected on guinea pig , human skin
Autoantibodies specific to Dsg1 are demonstrable by ELISA.

-
- Other forms of pemphigus
-
- Bullous impetigo: .
- Subcorneal pustular dermatosis (SCPD)
- Seborrheic dermatitis
- Subacute cutaneous lupus erythematosus

- Localized disease $\Rightarrow$ Superpotent topical corticosteroids
- - Active and widespread disease treatment is similar to that for pemphigus vulgaris (systemic therapy).
- - Dapsone can also be used when neutrophils are dominant histologically

Variants



Fig. 29.9 Pemphigus erythematosus. Erythematous plaques with scale-crust and erosions on the nose and malar area of the face. Courtesy, Ronald P Rapini, MD.

- 1- pemphigus erythematosus (senear usher)
 - Localized variant
 - Scaly crusted lesion at malar resion
 - Induced by UV
 - +ve ANA
 - If : 1- igG , c3 at KCS and BMZ
 - Lupus band test +ve
 - ANA +ve

Fogo selvagem (wild fire)



Endemic >>>brazil

Young adult , children

Rural , river

Black flies(simulim spp)

Bugs

Sun exposure

Paraneoplastic pemphigus

- PNP is a rare type of pemphigus (3-5 % of all pemphigus cases) associated with underlying neoplasms (whether malignant or benign)
- The most commonly associated neoplasms are...
 - ☆ Non-Hodgkin lymphoma (40%)
 - % Chronic lymphocytic leukemia (30%)
 - A- Castleman's disease (10%):
 - % Malignant and benign thymomas (6%)
 - V Sarcomas (6%)
 - Y Waldenstrom's macroglobulinemia (6%)

- Pathogenesis (target proteins)

Polymorphic - targets many antigens (BOTH in desmosome and BMZ).

- ☆ Desmoglein 3, 1

- ☆ Desmoplakin I, II

A Envoplakin / Periplakin: **vip**

- ☆ BPAG1 (bullous pemphigoid antigen 1)

- ☆ Plectin

c/p

- **Intractable severe stomatitis:**
 - Most common , earliest sign .
 - Extend to vermillion border , extremely resistant
- **Pseudo membranous conjunctivitis >>> scar**
- **Skin: polymorphic**
 1. Pemphigus vulgaris
 2. Bollous pemphigoid
 3. EM
 4. lp
- **systemic affection: bronchiolitis obliterans**, which can be fatal as a result of respiratory failure





DISORDERS WITH HEMORRHAGIC CRUSTS OF THE VERMILION LIPS

Herpes simplex
Herpes zoster
Erythema multiforme major
Stevens–Johnson syndrome/TEN spectrum
Pemphigus vulgaris
Paraneoplastic pemphigus
Contact cheilitis



Table 29.8 Disorders with hemorrhagic crusts of the vermilion lips.

Photograph, courtesy, Jeffrey P Callen, MD.

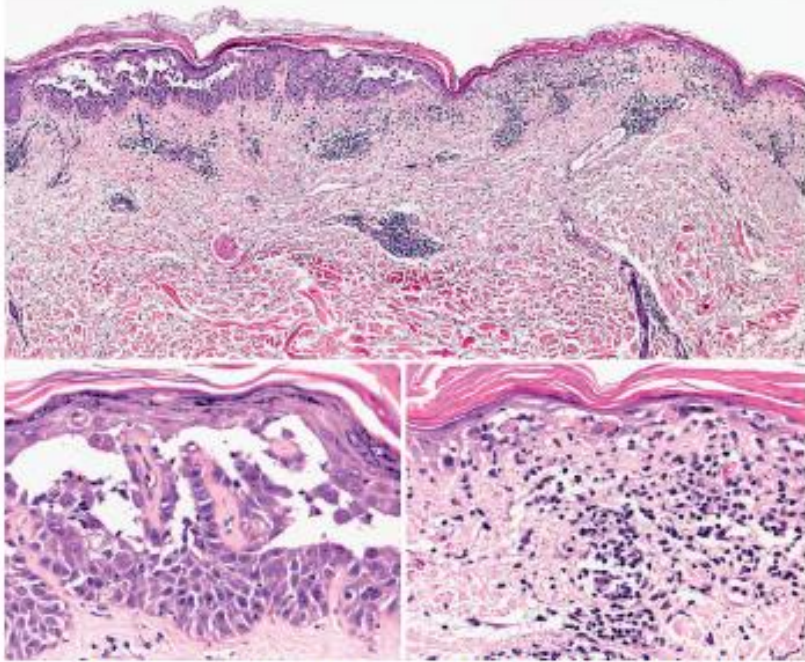


Fig. 29.15 Paraneoplastic pemphigus – histologic features. In the same biopsy specimen, there are areas of suprabasilar acantholysis (left inset) as well as an interface dermatitis with basal cell vacuolar change, necrotic keratinocytes, and lymphocytes within the epidermis (right inset). Courtesy, Lorenzo Centoní, MD.

- cutaneous lesions: (polymorphism)
 1. pemphigus vulgaris-like A supra-basilar acantholysis
 2. erythema multiforme-like ^ individual keratinocyte necrosis with lymphocytes within the epidermis
 3. lichen planus-like ■=>basal cell liquefactive degeneration or a band-like dense lymphocytic infiltrate in the upper dermis

- DIF: Intercellular and subepidermal IgG.
- IIF: rat bladder is used to detect paraneoplastic pemphigus
- treatment
 - Benign tumors (e.g. thymomas or localized Castleman's disease) :surgical excision.
 - Malignant neoplasms tumor-specific chemotherapy

TIP

Associations with Paraneoplastic Pemphigus

- Malignancy, especially non-Hodgkin's lymphoma (42%)
- Chronic lymphocytic leukemia (29%)
- Castleman's (6%)
- Sarcoma (6%)
- Thymoma (6%)

TIP

Criteria for Paraneoplastic Pemphigus: Need 3 Major or 2 Major Plus 2 Minor

- Major criteria
 - Polymorphous mucocutaneous eruption
 - Concurrent internal neoplasia
 - Characteristic serum immunoprecipitation findings
- Minor criteria
 - Positive IIF on rat bladder
 - Positive DIF
 - Acantholysis on biopsy

IgA pemphigus



- igA pemphigus is a special type of pemphigus characterized .
- 1. immunopathologically by the presence of IgA autoantibodies (rather than IgG).
 2. histologically by neutrophils infiltrate (rather than eosinophils).
 3. clinically by pus-filled blisters (i.e. pustules) in annular configuration with pruritus.
 4. therapeutically by response to dapsone

- IgA pemphigus is divided into 2 subtypes the skin (i.e. non-immune).
depending on the level of intra-epidermal
-
- 1. Subcorneal pustular dermatosis (SCPD)
-
- 2. Intra-epidermal neutrophilic (IEN) type: supra-basilar pustules

Pathogenesis :

-
- 1. SCPD type: desmocollin 1
-
- 2. IEN type: still unknown target antigen (?)..

Clinical features

- Good general condition
- Rare mucosa
- Pruritis
- Sun flower like in groin , axilla

- intra-epidermal pustule containing neutrophils
-
- SCPD type: pustules are located sub-corneally in the upper epidermis.
-
- IEN type: supra-basilar pustules involving the lower or entire epidermis.

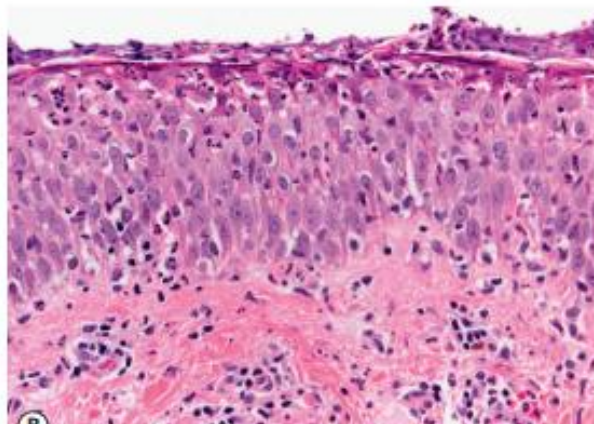
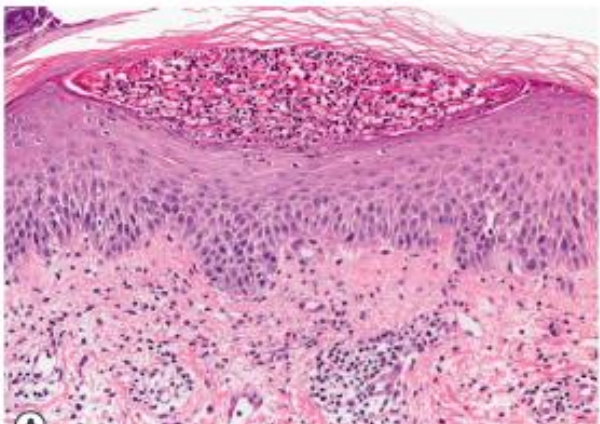


Fig. 29.16 IgA pemphigus – histologic features. Intraepidermal pustule with neutrophils present subcorneally in the subcorneal pustular dermatosis type (A), and within the entire epidermis in the intraepidermal neutrophilic type (B). Courtesy, Lorenzo Cerroni, MD.

- immunofluorescence
 - Direct IF: IgA deposition on cell surfaces of epidermal keratinocytes.
 - Indirect IF: circulating IgA autoantibodies.
- Treatment :
 - **Dapsone**
 - is the drug of choice MCQ (it suppresses functions of neutrophils) Dose: 25-125 mg/day.
 - sulfapyridine
 - low- to medium-dose prednisone, photochemotherapy (PUVA)
 - colchicine

Drug induced pemphigus

- Thiol group :
 - Sulpha group
 - Captopril
- Non thiol
 - Ace inh
 - Angioten-2 receptors blocker
 - Ca channel blocker

Pemphigus foliaceus

Pem erythematosus

Remission after offending drug

- **Drug-induced:** furosemide, NSAIDs, phenacetin, PCN-derivatives, gold, potassium iodide, captopril, enalapril, D-penicillamine, sulfasalazine

Drug-induced: **PF ChaNGS** Penicillamine **PCN-derivatives** Phototherapy
Furosemide Captopril NSAID Gold Sulfasalazine

Anti-p105 pemphigoid (105 kDa) } Similar to BP but autoantibodies targeting
Anti-p200 pemphigoid (200 kDa) } distinct antigens of epidermal BMZ

Immunofluorescence (IF) techniques:

Direct IF (DIF)	Detects <i>in vivo</i> antibodies bound to tissue antigens in perilesional skin
Indirect IF (IIF)	Detects circulating serum antibodies (substrate sections react with serially diluted serum from patient → incubated with anti-IgG or other specific fluorescent dye-tagged antibody)
	Best substrate: monkey esophagus (PV) , guinea pig esophagus (PF) , transitional rat bladder epithelium (PNP)
Salt-split-skin (SSS) technique	Variant of IIF allowing distinction between different subepidermal blistering conditions with similar DIF findings ; normal human skin incubated in 1M NaCl for 48-72 hrs resulting in split at lamina lucida level; location of antibody binding to split (epidermal or dermal side) distinguishes different diseases

Location of Split	Disease Examples
Subcorneal	Pemphigus foliaceus Pemphigus erythematosis Staphylococcal scalded skin syndrome Bullous impetigo
Suprabasilar	Pemphigus vulgaris
At/within basal layer	Bullous pemphigoid Epidermolysis bullosa simplex
Below basal keratinocytes	Bullous pemphigoid Linear IgA disease Cicatricial/mucous membrane pemphigoid Porphyria cutanea tarda Bullous lupus erythematosus Epidermolysis bullosa acquisita Dermatitis herpetiformis

TABLE 2.3.2 DISORDERS ASSOCIATED WITH ARMADILLO PROTEINS

Mutations: Armadillo	Disorder
β -Catenin	Some pilomatricomas Colorectal carcinoma
Plakoglobin	Naxos disease
Plakophilin 1	Ectodermal dysplasia with skin fragility
Antigenic: Armadillo	Disorder
Plakoglobin	Coprecipitates with desmogleins in pemphigus foliaceus, pemphigus vulgaris, and paraneoplastic pemphigus

Intermediate Filaments

- Found within a variety of cells, many different types
 - Type I = acidic keratins 9-20, chromosome 17
 - Type II = basic keratins 1-8, chromosome 12
 - Type III = vimentin, glial fibrillary acidic protein (GFAP), desmin, peripherin
 - Type IV = neurofilaments
 - Type V = nuclear lamins
 - Type VI = nestin

TABLE 2.3.5 DISORDERS ASSOCIATED WITH KERATINS

Mutations: Keratin	Disorder
Keratin 1	Ichthyosis hystrix
Keratins 1/10	Epidermolytic ichthyosis Epidermal nevi with EHK
Keratins 1/16	Nonepidermolytic PPK
Keratin 2e	Ichthyosis bullosa of Siemens
Keratins 3/12	Corneal dystrophy of Meesmann
Keratins 4/13	White sponge nevus of Cannon
Keratins 5/14	Epidermolysis bullosa simplex variants
Keratins 6a/16	Pachyonychia congenita
Keratins 6b/17	Pachyonychia congenita
Keratin 9	Epidermolytic PPK
Lamin A	Progeria

TABLE 2.3.3 DISORDERS ASSOCIATED WITH CADHERIN PROTEINS

Mutations: Cadherin	Disorder
Desmoglein 1	Striate palmoplantar keratoderma (PPK) type 1
E-cadherin	Gastric cancer
Antigenic: Cadherin	Disorder
Desmoglein 1	Pemphigus foliaceus Staphylococcal scalded skin syndrome (toxin mediated)
Desmogleins 1/3	Mucocutaneous pemphigus vulgaris
Desmoglein 3	Mucosal pemphigus vulgaris
Desmocollin	IgA pemphigus

TABLE 2.3.4 DISORDERS ASSOCIATED WITH INTEGRINS

Mutations: Integrin	Disorder
$\alpha_6\beta_4$ -Integrin	Junctional epidermolysis bullosa (JEB) with pyloric atresia
Antigenic: Integrin	Disorder
β_4 -Integrin	Ocular cicatricial pemphigoid

TABLE 2.3.7 DISORDERS ASSOCIATED WITH CONNEXINS

Mutations: Connexins	Disorder
26	Vohwinkel syndrome Keratitis-ichthyosis-deafness (KID) syndrome Palmoplantar keratoderma (PPK) with deafness
30	Hidrotic ectodermal dysplasia
30.3, 31	Erythrokeratoderma variabilis

TABLE 2.3.8 DISORDERS ASSOCIATED WITH PROTEINS OF THE HEMIDESMOSOME

Mutations: Protein	Disorder
BPAg2	JEB, non-Herlitz type
Laminin V	JEB, Herlitz type
Plectin	Epidermolysis bullosa simplex with muscular dystrophy
$\alpha_6\beta_4$ -Integrin	Junctional epidermolysis bullosa with pyloric atresia
Antigenic: Protein	Disorder
BPAg1 and BPAg2-NC16A	Bullous pemphigoid
BPAg2 97-kDa antigen	Lichen planus pemphigoides Linear IgA disease
BPAg2-NC16A	Herpes gestationis
BPAg2-C-terminal domain	Mucous membrane pemphigoid
Laminin 5	Mucous membrane pemphigoid
β_4 -Integrin	Ocular cicatricial pemphigoid

Thank you